Ethan Collins

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EDUCATION

The University of Texas at Dallas, Richardson, TX — Expected May 2028
Bachelor of Science, Software Engineering

TECHNICAL PROJECTS

Circuit OS (Python, Pandas, scikit-learn, Flask) — Sports Performance Analytics Engine — October 2025

- Built a real-time sports performance prediction engine by training a RandomForest regression model on dynamic athlete variables and live environmental conditions, replacing a fully manual interval analysis process.
- Enabled data-driven athlete performance segmentation by engineering a time-series feature extraction pipeline in NumPy that automatically computed behavioral metrics across entire datasets, eliminating manual tabulation.
- Ensured high-fidelity data quality across the full analytics pipeline by implementing TDD with Python's unittest framework, catching and resolving edge cases before they could corrupt downstream predictive models.
- Delivered real-time performance analytics to live reporting dashboards by building a Flask REST API that automatically computed and served probability scores and key performance indicators on each incoming data point.

EarlyAxxess (Neo4j, Python, FastAPI, Celery, Redis) — GraphRAG Medical Reasoning System — February 2026

- Delivered sub-second diagnostic reasoning across a 6M-node medical knowledge graph by orchestrating 3 distributed Celery worker pools (NLP, Sync, Reasoning) with Redis message brokering and Bayesian-weighted Dijkstra pathfinding via NetworkX.
- Built a counterfactual What-If simulator that clones the reasoning subgraph, removes causal nodes, and re-runs Dijkstra pathfinding to quantify health score delta and explain symptom-causation changes in under 2 seconds.
- Reduced erroneous clinical outputs to near-zero by automating LLM validation through a fuzzy entity-matching pipeline, with safe-template fallback preventing unverified inferences from reaching the output layer.

CometNavigator (Next.js, TypeScript, MongoDB Atlas) — AI-Powered Student Dashboard — March 2026

- Enabled semantic student-resource matching for 25,000+ users by generating 384-dimensional embeddings with sentence-transformers (all-MiniLM-L6-v2) and scoring profiles via cosine similarity.
- Achieved sub-200ms query latency by caching vector embeddings in MongoDB Atlas and optimizing FastAPI endpoints for low-overhead retrieval.

RateMyCareer (Python, Flask, React.js, Supabase) — Career Sentiment Analysis Platform — December 2025

- Eliminated manual data processing overhead across 10,000+ records by engineering an automated ETL pipeline to extract, preprocess, and load social media posts with real-time updates.
- Improved sentiment classification accuracy by 45% by redesigning the ML inference layer, replacing a rule-based VADER system with a fine-tuned RoBERTa pipeline optimized for tokenization and preprocessing.
- Reduced time-to-insight by architecting a modular REST API in Flask that exposed ML predictions directly to the React.js dashboard, removing the manual reporting step between data ingestion and display.

TECHNICAL SKILLS

Languages: Java, Python, C++, PostgreSQL, HTML5, CSS, React.js, Next.js, Node.js, Flask, FastAPI

Artificial Intelligence/Machine Learning: TensorFlow, scikit-learn, RoBERTa, OpenCV, NumPy, Pandas

Tools: Supabase, Neo4j, MongoDB, AWS (EC2), Redis, NetworkX, Docker, Celery, Excel, Word, Snowflake Cortex

Methodologies: Agile, TDD, ETL, process improvement

Spoken Languages: English (Native), Japanese (JLPT N4 In-Progress)

LEADERSHIP & EXPERIENCE

Project AI Implementation Lead (RateMyCareer) — Artificial Intelligence Mentorship (UTD Club), Fall 2025

- Directed the technical architecture on a project for a team of 6 by selecting core machine learning models and frameworks, coordinating cross-functional database schema design and frontend visualization.
- Presented the end-to-end AI framework and development lifecycle to a panel of industry experts from leading technology companies (including Google and Apple), successfully defending model selection and integration strategies.

Team Captain — FTC Team 8626 (Innovate & Design Award Winner), Spring 2025

HONORS & AWARDS

- **DFW Stock Competition Winner**, STREAM Foundation (106% ROI — $2.1M 2-month simulated portfolio)
- **National African American Recognition Award**, College Board
- **2nd Place Advanced League CodeQuest Winner**, Lockheed Martin
- **Figma Design Track Winner**, HackAI Hackathon